

```
// Functional description of JK flip-flop using a case  
statement
```

```
module JK_FF (input J, K, Clk, output reg Q, output  
Q_b);
```

```
    assign Bib = ~Q;
```

```
    always @ (posedge Clk)
```

```
    case ({J,K})
```

```
        2'b00: Q <= Q;
```

```
        2'b01: Q <= 1'b0;
```

```
        2'b10: Q <= 1'b1;
```

```
        2'b11: Q <= !Q;
```

```
    encase;
```

```
endmodule;
```